SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool : Debugging & Optimisation Task (Low)
Assessment Tool Weightage: 30%

Dr. Hemavathi R

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Identify and fix the error in the given SQL query that uses incorrect JOIN syntax and returns duplicate results.	BL3 – Apply	**5**
2	The following sub-query returns incorrect results due to a missing correlation. Debug and rewrite it correctly.	BL3 – Apply	**5**
3	A given SQL view definition causes errors when queried. Identify the issue and rewrite a correct and optimized view.	BL3 – Apply	**5**
4	Debug the given relational algebra expression that produces incorrect output for a natural join operation.	BL3 – Apply	**5**
5	The SQL query below violates referential integrity. Identify the violation and modify the statements to enforce proper constraints.	BL3 – Apply	**5**
6	Given an inefficient SQL query with nested loops, rewrite it using JOIN to optimize performance.	BL6 – Create	**5**
7	The GROUP BY query below produces wrong aggregation results. Debug the query and explain the error.	BL3 – Apply	**5**
8	Identify and fix the logical error in a given SQL UPDATE statement that unintentionally modifies all rows in the table.	BL3 – Apply	**5**

9	Optimize the given multi-table SQL query by restructuring sub-queries into JOINs and using appropriate indexes.	BL6 – Create	**5**
10	Debug a given relational calculus expression that returns an empty result set due to incorrect quantifier usage.	BL3 – Apply	**5**

Code of Conduct:

I Sree (192524023) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Sree B

1.Identify and fix the error in the given SQL query that uses incorrect JOIN syntax and returns duplicate results.

```
mysql> SELECT * FROM Student;
+------------+-------+------+---------------+
| student_id | name  | age  | Department_id |
+------------+-------+------+---------------+
|        101 | Rahul |   20 |           101 |
|        102 | Priya |   21 |           102 |
|        103 | Arun  |   22 |           103 |
|        104 | Kiran |   20 |           104 |
|        105 | Neha  |   21 |           105 |
+------------+-------+------+---------------+
5 rows in set (0.010 sec)
```

2.The following sub-query returns incorrect results due to a missing correlation. Debug and rewrite it correctly.

```
mysql> CREATE TABLE Employee (
    ->     emp_id INT PRIMARY KEY,
    ->     emp_name VARCHAR(30),
    ->     salary INT,
    ->     Department_id INT,
    ->     FOREIGN KEY (Department_id) REFERENCES Department(Department_id)
    -> );
Query OK, 0 rows affected (1.288 sec)

mysql> INSERT INTO Employee VALUES
    -> (1,'Rahul',40000,101),
    -> (2,'Priya',50000,101),
    -> (3,'Arun',35000,102),
    -> (4,'Neha',60000,102),
    -> (5,'Kiran',45000,103);
Query OK, 5 rows affected (0.207 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Employee;
+--------+----------+--------+---------------+
| emp_id | emp_name | salary | Department_id |
+--------+----------+--------+---------------+
|      1 | Rahul    |  40000 |           101 |
|      2 | Priya    |  50000 |           101 |
|      3 | Arun     |  35000 |           102 |
|      4 | Neha     |  60000 |           102 |
|      5 | Kiran    |  45000 |           103 |
+--------+----------+--------+---------------+
5 rows in set (0.034 sec)
```

3.A given SQL view definition causes errors when queried. Identify the issue and rewrite a correct and optimized view.

```
mysql> SELECT * FROM Department_View;
+---------------+----------------+-----------+
| Department_id | Departmem_name | block     |
+---------------+----------------+-----------+
|           101 | CSE            | AHS       |
|           102 | ECE            | ECE Block |
|           103 | Civil          | SCAD      |
|           104 | IT             | ADMIN     |
|           105 | Bio-Tech       | RHS       |
+---------------+----------------+-----------+
5 rows in set (0.044 sec)

mysql> CREATE OR REPLACE VIEW Department_View AS
    -> SELECT
    ->     Department_id,
    ->     Departmem_name,
    ->     block
    -> FROM Department;
Query OK, 0 rows affected (0.102 sec)

mysql> SELECT * FROM Department_View;
+---------------+----------------+-----------+
| Department_id | Departmem_name | block     |
+---------------+----------------+-----------+
|           101 | CSE            | AHS       |
|           102 | ECE            | ECE Block |
|           103 | Civil          | SCAD      |
|           104 | IT             | ADMIN     |
|           105 | Bio-Tech       | RHS       |
+---------------+----------------+-----------+
5 rows in set (0.015 sec)
```

4. Debug the given relational algebra expression that produces incorrect output for a natural join operation.

```
mysql> SELECT e.student_id,
    ->        c.course_id,
    ->        c.course_name
    -> FROM Enroll e
    -> JOIN Course c
    -> ON e.course_id = c.course_id;
+------------+-----------+-------------+
| student_id | course_id | course_name |
+------------+-----------+-------------+
|        101 |         1 | DBMS        |
|        101 |         2 | Java        |
|        102 |         2 | Java        |
|        103 |         3 | Python      |
+------------+-----------+-------------+
4 rows in set (0.377 sec)
```

5. The SQL query below violates referential integrity. Identify the violation and modify the statements to enforce proper constraints

```
mysql> SELECT * FROM Employee;
+--------+----------+--------+---------------+
| emp_id | emp_name | salary | Department_id |
+--------+----------+--------+---------------+
|      1 | Rahul    |  40000 |           101 |
|      2 | Priya    |  50000 |           101 |
|      3 | Arun     |  35000 |           102 |
|      4 | Neha     |  60000 |           102 |
|      5 | Kiran    |  45000 |           103 |
|      6 | Rahul    |  40000 |           110 |
+--------+----------+--------+---------------+
6 rows in set (0.009 sec)

mysql> SELECT * FROM Department;
+---------------+----------------+-----------------+---------+-----------+
| Department_id | Departmem_name | numberofcourses | credits | block     |
+---------------+----------------+-----------------+---------+-----------+
|           101 | CSE            |              32 |     192 | AHS       |
|           102 | ECE            |              32 |     192 | ECE Block |
|           103 | Civil          |              30 |     190 | SCAD      |
|           104 | IT             |              32 |     192 | ADMIN     |
|           105 | Bio-Tech       |              32 |     192 | RHS       |
|           110 | Mechanical     |              30 |     180 | ME Block  |
+---------------+----------------+-----------------+---------+-----------+
6 rows in set (0.010 sec)

mysql>
```

6. Given an inefficient SQL query with nested loops, rewrite it using JOIN to optimize performance

```
mysql> SELECT DISTINCT
    ->        c.customer_id,
    ->        c.customer_name,
    ->        c.city
    -> FROM Customer c
    -> JOIN Orders o
    -> ON c.customer_id = o.customer_id;
+-------------+---------------+-----------+
| customer_id | customer_name | city      |
+-------------+---------------+-----------+
|           1 | Rahul         | Chennai   |
|           2 | Priya         | Bangalore |
|           4 | Neha          | Delhi     |
+-------------+---------------+-----------+
3 rows in set (0.398 sec)

mysql>
```

7. The GROUP BY query below produces wrong aggregation results. Debug the query and explain the error.

```
mysql> SELECT product_name,
    ->        SUM(quantity) AS Total_Quantity
    -> FROM Sales
    -> GROUP BY product_name;
+--------------+----------------+
| product_name | Total_Quantity |
+--------------+----------------+
| Pen          |             30 |
| Book         |             25 |
| Pencil       |             25 |
+--------------+----------------+
3 rows in set (0.011 sec)
```

8. Identify and fix the logical error in a given SQL UPDATE statement that unintentionally modifies all rows in the table

```
mysql> SELECT * FROM Student;
+------------+-------+------+---------------+
| student_id | name  | age  | Department_id |
+------------+-------+------+---------------+
|        101 | Rahul |   23 |           101 |
|        102 | Priya |   21 |           102 |
|        103 | Arun  |   22 |           103 |
|        104 | Kiran |   20 |           104 |
|        105 | Neha  |   21 |           105 |
+------------+-------+------+---------------+
5 rows in set (0.009 sec)
```

9. Optimize the given multi-table SQL query by restructuring sub-queries into JOINs and using appropriate indexes.

```
mysql> SHOW INDEX FROM Orders;
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+------
---------+---------+------------+
| Table  | Non_unique | Key_name           | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index
_comment | Visible | Expression |
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+------
---------+---------+------------+
| orders |          0 | PRIMARY            |            1 | order_id    | A         |           4 |     NULL |   NULL |      | BTREE      |         |
         | YES     | NULL       |
| orders |          1 | idx_orders_customer |            1 | customer_id | A         |           3 |     NULL |   NULL | YES  | BTREE      |         |
         | YES     | NULL       |
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+------
---------+---------+------------+
2 rows in set (0.546 sec)
```

```
mysql> SHOW INDEX FROM Payment;
+---------+------------+-----------------+--------------+------------+-----------+-------------+----------+--------+------+------------+---------+---------------+
| Table   | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+---------+------------+-----------------+--------------+------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| payment |          0 | PRIMARY         |            1 | payment_id | A         |           3 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| payment |          1 | idx_payment_order |            1 | order_id   | A         |           3 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+---------+------------+-----------------+--------------+------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
2 rows in set (0.182 sec)
```

10. Debug a given relational calculus expression that returns an empty result set due to incorrect quantifier usage.

```
mysql> SELECT * FROM Student;
+------------+-------+------+---------------+
| student_id | name  | age  | Department_id |
+------------+-------+------+---------------+
|        101 | Rahul |   23 |           101 |
|        102 | Priya |   21 |           102 |
|        103 | Arun  |   22 |           103 |
|        104 | Kiran |   20 |           104 |
|        105 | Neha  |   21 |           105 |
+------------+-------+------+---------------+
5 rows in set (0.008 sec)

mysql> SELECT s.student_id,
    ->        s.name,
    ->        d.Departmem_name
    -> FROM Student s
    -> JOIN Department d
    -> ON s.Department_id = d.Department_id;
+------------+-------+----------------+
| student_id | name  | Departmem_name |
+------------+-------+----------------+
|        101 | Rahul | CSE            |
|        102 | Priya | ECE            |
|        103 | Arun  | Civil          |
|        104 | Kiran | IT             |
|        105 | Neha  | Bio-Tech       |
+------------+-------+----------------+
5 rows in set (0.011 sec)

mysql>
```